

9.5.7.

A computer programming team has 13 members.

- a. How many ways can a group of seven be chosen to work on a project?
- b. Suppose seven team members are women and six are men.
 - (i) How many groups of seven can be chosen that contain four women and three men?
 - (ii) How many groups of seven can be chosen that contain at least one man?
 - (iii) How many groups of seven can be chosen that contain at most three women?
- c. Suppose two team members refuse to work together on projects. How many groups of seven can be chosen to work on a project?
- d. Suppose two team members insist on either working together or not at all on projects. How many groups of seven can be chosen to work on a project?

- a. How many ways can a group of seven be chosen to work on a project?

Since the order of the groups chosen to work on a project does not matter, then:

$$\begin{aligned}
 \binom{13}{7} &= \frac{13!}{7!(13-7)!} = \frac{13!}{7! \times 6!} \\
 \frac{13!}{7! \times 6!} &= \frac{13 \times 12 \times 11 \times 10 \times 9 \times 8 \times 7!}{7! \times 6!} = \\
 \frac{13 \times 12 \times 11 \times 10 \times 9 \times 8 \times 7!}{7! \times 6!} &= \frac{13 \times 12 \times 11 \times 10 \times 9 \times 8}{6!} = \\
 \frac{13 \times 12 \times 11 \times 10 \times 9 \times 8}{6!} &= \frac{1,235,520}{720} = 1,716
 \end{aligned}$$

Solution: a. There're 1,716 ways to form a group of seven to work on a project with 13 members.

- b. Suppose seven team members are women and six are men.
 - (i) How many groups of seven can be chosen that contain four women and three men?

We know that there're seven women and six men. And we must show how many groups of seven can be chosen which contain only four women and three men. These are our known constraints, and since the order of the groups chosen to work on a project does not matter, then:

$$\binom{7}{4} \times \binom{6}{3}$$

$$\binom{7}{4} = \frac{7!}{4!(7-4)!} = \frac{7!}{4! \times 3!} = \frac{7 \times 6 \times 5 \times 4!}{4! \times 3!} = \frac{7 \times 6 \times 5}{3!} = \frac{210}{6} = \boxed{35}$$

$$\binom{6}{3} = \frac{6!}{3!(6-3)!} = \frac{6!}{3! \times 3!} = \frac{6 \times 5 \times 4 \times 3!}{3! \times 3!} = \frac{6 \times 5 \times 4}{3!} = \frac{120}{6} = \boxed{20}$$

$$35 \times 20 = \boxed{700}$$

Solution: b. (i) There're 700 possible groups containing exactly four women and three men.

b. Suppose seven team members are women and six are men.

(ii) How many groups of seven can be chosen that contain at least one man?

We know that there're seven women and six men. And we must show how many groups of seven can be chosen which contain at least one man. There is only one group which violates this constraint, which is the group containing only women, therefore:

$$\binom{13}{7} - \binom{7}{7}$$

$$\binom{13}{7} = 1,716$$

$$\binom{7}{7} = 1$$

$$1,716 - 1 = \boxed{1,715}$$

Solution: b. (ii) There're 1,715 groups that contain at least one man.

b. Suppose seven team members are women and six are men.

(iii) How many groups of seven can be chosen that contain at most three women?

We know that there're seven women and six men. And we must show how many groups of seven can be chosen which contain at most three women. These are our known constraints, which means that we calculate all forbidden cases, and subtract them from the total cases. Hence:

$$\binom{13}{7} - \sum_{w=4}^7 \left(\binom{7}{w} \times \binom{6}{7-w} \right)$$

$$\binom{13}{7} = 1,716$$

$$\binom{7}{4} \binom{6}{3} = 35 \times 20 = 700$$

$$\binom{7}{5} \binom{6}{2} = 21 \times 15 = 315$$

$$\binom{7}{6} \binom{6}{1} = 7 \times 6 = 42$$

$$\binom{7}{7} \binom{6}{0} = 1 \times 1 = 1$$

$$700 + 315 + 42 + 1 = 1,058$$

$$1,716 - 1,058 = \boxed{658}$$

Solution: b. (iii) There're 658 groups containing at most three women.

c. Suppose two team members refuse to work together on projects. How many groups of seven can be chosen to work on a project?

We know that there are 13 members, and two team members refuse to work together on projects. We must show how many groups of seven can be chosen without these two members working together. We therefore calculate the number of groups containing both members and subtract these from the total number of groups. If both members are included, then five additional members must be chosen from the remaining 11 members. Hence:

$$\binom{13}{7} - \binom{11}{5}$$

$$\begin{aligned}\binom{13}{7} &= \boxed{1,716} \\ \binom{11}{5} &= \frac{11!}{5!(11-5)!} = \frac{11!}{5! \times 6!} = \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6!}{5! \times 6!} = \\ &= \frac{11 \times 10 \times 9 \times 8 \times 7}{5!} = \frac{55,440}{120} = \boxed{462} \\ 1,716 - 462 &= \boxed{1,254}\end{aligned}$$

Solution: c. There're 1,254 groups of seven where the two members are not working together.

d. Suppose two team members insist on either working together or not at all on projects. How many groups of seven can be chosen to work on a project?

We know that there are 13 members, and two team members insist on either working together or not at all. There are therefore two possible cases. If both members are included, five additional members must be chosen from the remaining 11 members. If neither member is included, seven members must be chosen from the remaining 11 members. Hence:

$$\binom{11}{5} + \binom{11}{7} = 462 + 330 = \boxed{792}$$

Solution: d. There're 792 groups of seven where the two members either work together or not at all.